

Mineral Dissolution and Precipitation Reactions

The rate law for the mineral dissolution and precipitation reaction Equation (D5.2) is (Bethke, 1996):

$$r_j = \hat{A}_j k_j \left(1 - \frac{Q_j}{K_{eq,j}} \right), \quad j = 1, \dots, R_{mn} \quad (D8.1)$$

where r_j is the rate, $\hat{A}_j$ is the reactive surface area for mineral j , k_j is the rate constant of mineral reaction j , $K_{eq,j}$ is the chemical equilibrium constant for mineral reaction j and Q_j is the activity product of mineral reaction j . The activity product Q_j is analogous to the activity product for aqueous chemical equilibrium reactions Equation (D6.2), i.e.

$$Q_j = \prod_{i=1}^{n_{aq}} a_i^{v_{ij}} \quad (D8.2)$$

The product does not involve the activities of the minerals as they are equal to unity. The chemical equilibrium constants $K_{eq,j}$ for many minerals are also available in the literature (Kharaka et al, 1989, and Delany and Lundeen, 1990). The ratio $Q_j / K_{eq,j}$ is called the *saturation index* of the reaction. If $Q_j / K_{eq,j} > 1$, mineral precipitation occurs and if $Q_j / K_{eq,j} < 1$, mineral dissolution occurs. In this document, the convention adopted is that the rate r_j is negative for dissolution and positive for precipitation. The rate is zero when $Q_j / K_{eq,j} = 1$.

The rate of dissolution/precipitation in Equation (D8.1) applies to the mineral. The rate of formation/consumption of the different aqueous species is obtained by multiplying r_j by the respective stoichiometry coefficient:

$$r_{ij} = v_{ij} \cdot r_j \quad (D8.3)$$

The reaction rate constants, k_j , are normally reported in the literature at a reference temperature T_0 (usually 298.15 K or 25 °C). The following equation is used to calculate the rate constant at a different temperature T :

$$k_j = k_{0j} \exp \left[-\frac{E_{aj}}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right) \right] \quad (D8.4)$$

where E_{aj} is the activation energy for reaction j [J/mol] and k_{0j} is the reaction rate constant for reaction j at T_0 [mol/(m² s)].

The reactive surface area $\hat{A}_j$ is another important parameter in the calculation of the rate. The following equation is used to calculate the reactive surface area with change in the moles of minerals through dissolution or precipitation:

$$\hat{A}_j = \hat{A}_j^0 \cdot \frac{N_j}{N_j^0} \quad (\text{D8.5})$$

where $\hat{A}_j^0$ is the reactive surface area at time 0, N_j is the mole number of mineral j per unit grid block volume at current time and N_j^0 is the mole number of mineral j per unit grid block bulk volume at time 0.

Mineral dissolution and precipitation change the void volume of the porous medium. The porosity is calculated as follows:

$$\hat{\phi}^* \equiv \phi^* - \sum_{j=1}^{n_m} \left(\frac{N_j}{\rho_j} - \frac{N_j^0}{\rho_j} \right) \quad (\text{D8.6})$$

$$\phi = \hat{\phi}^* [1 + c_\phi (p - p^*)] \quad (\text{D8.7})$$

where ϕ is the porosity, ϕ^* is the reference porosity without mineral precipitation/dissolution, $\hat{\phi}^*$ is the reference porosity including mineral precipitation/dissolution, N_j is the total moles of mineral j per bulk volume at the current time, N_j^0 is the total moles of mineral j per bulk volume at time 0, ρ_j is the mineral molar density and c_ϕ is the rock compressibility, p^* is the reference pressure.

The absolute permeability also changes with mineral dissolution and precipitation. The Kozeny-Carman equation is used:

$$\frac{k}{k^0} = \left(\frac{\phi}{\phi^0} \right)^3 \cdot \left(\frac{1 - \phi^0}{1 - \phi} \right)^2 \quad (\text{D8.8})$$

where k^0 and ϕ^0 are the initial permeability and porosity, respectively.